

ECE 222 In-Class Problems 2

Instruction:

In this discussion, we want to give you more practice with Coulomb's law and Gauss's law, think about when to use each, and help you become comfortable with setting up these problems. We'll also solve an integral that will show up repeatedly in this class.

At the end of class, take a photograph(s) or screenshot(s) of your answers and upload to today's in-class problem assignment on Canvas.

See my work on Page 3

Problem 1 (similar to 4.16 in Homework):

A line segment of length L is uniformly charged with line charge density ρ_l (ρ_l is positive) and placed along the z -axis between $z = -L/2$ and $z = L/2$. Calculate $\vec{E}$ in the x - y plane.

a) Start by making a sketch of the problem, then think about, discuss these questions with your neighbor(s), and write down responses:

- What coordinate system should you use? Why?
- In what direction do you expect $\vec{E}$ in the x - y plane to point? Why?
- Should you use Coulomb's law or Gauss's law to solve this problem? Why?

b) Begin working the problem until you get to an equation of this form:

$$\vec{E} = \hat{r} \int_{-L/2}^{L/2} [\dots] dz_0 + \hat{z} \int_{-L/2}^{L/2} [\dots] dz_0$$

Based on your responses to (a), do you actually need to evaluate both of the terms in this integral? In other words, do you expect one of these terms to be equal to zero? If you have time at the end, show mathematically that this term will equal zero.

c) Evaluate the integral to find $\vec{E}$.

Hint:

How to $\int_{z=-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{(r^2+z^2)^{\frac{3}{2}}} dz$

1. Trigonometric Substitution


 $\text{let } \frac{z}{r} = \tan u \quad z = r \tan u \quad dz = r \frac{d(\tan u)}{du} = \frac{r}{\cos^2 u} du$
 $z \in [-\frac{L}{2}, \frac{L}{2}] \text{ becomes } u \in [-\tan^{-1}(\frac{L}{2r}), \tan^{-1}(\frac{L}{2r})]$

2. Simplify the denominator

$$(r^2+z^2)^{\frac{3}{2}} = \left[r^2 \left(1 + \left(\frac{z}{r} \right)^2 \right) \right]^{\frac{3}{2}} = r^3 \underbrace{\left(1 + \tan^2 u \right)^{\frac{3}{2}}}_{\substack{= \\ \frac{1}{\cos^2 u}}} = r^3 \frac{1}{\cos^3 u}$$

3. Transform the integral and evaluate

$$\int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{(r^2+z^2)^{\frac{3}{2}}} dz = \int_{-\tan^{-1}(\frac{L}{2r})}^{\tan^{-1}(\frac{L}{2r})} \left[\frac{\cos^3 u}{r^3} \cdot \frac{r}{\cos^2 u} \right] du \Rightarrow \frac{\cos u}{r^2}$$

$$= \frac{1}{r^2} \sin u \Big|_{-\tan^{-1}(\frac{L}{2r})}^{\tan^{-1}(\frac{L}{2r})}$$

$$= \frac{2}{r^2} \frac{\frac{L}{2r}}{\sqrt{1 + (\frac{L}{2r})^2}}$$

$$\begin{aligned} \sin u &= \tan u \cdot \cos u \\ &= \frac{\tan u}{\sqrt{1 + \tan^2 u}} \end{aligned}$$

If you don't finish this, don't worry! Review the solution later and make sure you can work through all the steps.

d) What does your result become as the line segment becomes very long (i.e. $L \rightarrow \infty$)?

Problem 2 : Consider the same problem as Problem 1, but instead assume the line is infinite ($L \rightarrow \infty$).

a) Should you use Coulomb's law or Gauss's law to solve this problem? Why?

b) Calculate $\vec{E}$ everywhere.

c) Which approach was easier (with Problem 1 or with Problem 2)? Note that Problem 1 with $L \rightarrow \infty$ would have required the exact same approach, just with integration limits of $\pm\infty$ instead of $\pm L/2$.

Problem 1:

a.) Cylindrical coordinates are best because the charge is aligned with the z -axis



- Radially outward it points if I imagine
- The endpoints destroy symmetry, so Coulomb's law should be used

$$\therefore d\vec{E} = \frac{1}{4\pi\epsilon} \frac{dq\hat{R}}{R^3} = \frac{\rho}{4\pi\epsilon} \frac{(r\hat{r} - z_0\hat{z})}{(r^2 + z_0^2)^{3/2}} dz_0$$

$$\vec{E} = \frac{\rho}{4\pi\epsilon} \int_{-L/2}^{L/2} \frac{r\hat{r} - z_0\hat{z}}{(r^2 + z_0^2)^{3/2}} dz_0$$

$$\vec{E} = \frac{\rho}{4\pi\epsilon} \left(\hat{r} \int_{-L/2}^{L/2} \frac{dz_0}{(r^2 + z_0^2)^{3/2}} - \hat{z} \int_{-L/2}^{L/2} \frac{z_0 dz_0}{(r^2 + z_0^2)^{3/2}} \right)$$

$$\vec{E} = \hat{r} \frac{\rho}{4\pi\epsilon} \int_{-L/2}^{L/2} \frac{dz_0}{(r^2 + z_0^2)^{3/2}} = \hat{r} \frac{\rho}{4\pi\epsilon} \left[\frac{z_0}{r^2 \sqrt{r^2 + z_0^2}} \right]_{-L/2}^{L/2}$$

$$\vec{E} = \hat{r} \frac{\rho L}{4\pi\epsilon \sqrt{r^2 + (L/2)^2}} = \hat{r} \frac{\rho L}{2\pi\epsilon \sqrt{4r^2 + L^2}}$$

$$\therefore \frac{L}{\sqrt{4r^2 + L^2}} = \frac{1}{L \sqrt{4r^2/L^2 + 1}} = \frac{1}{\sqrt{4r^2/L^2 + 1}} = \frac{1}{\sqrt{1 + 4r^2/L^2}}$$

$$\lim_{L \rightarrow \infty} \frac{1}{\sqrt{1 + 4r^2/L^2}} = 1, \quad \lim_{L \rightarrow \infty} \vec{E} = \frac{\rho}{2\pi\epsilon r} \hat{r}$$

Problem 2.) a. Gauss's law, symmetry is guaranteed now

$$\Phi_E = E(r)(2\pi r L)$$

$$\therefore \vec{E} \cdot d\vec{S} = 0; Q_{enc} = \rho L$$

$$E(2\pi r L) = \frac{\rho L}{\epsilon}; \quad \vec{E}(r) = \frac{\rho}{2\pi\epsilon r} \hat{r} \quad (r > 0)$$

Gauss is easier but requires symmetry